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TASK 3: DATA VISUALIZATION

1. Introduction

Data visualization is the process of presenting data through charts, graphs, and dashboards to make information easier to understand. In this task, Python and Matplotlib were used to visualize the Books Dataset created during Task 1 and analyzed during Task 2. Different visualizations were developed to understand book prices, price categories, ratings, and the relationship between prices and ratings.

2. Objectives

- To transform raw data into meaningful visual representations.
- To create charts using Python and Matplotlib.
- To identify patterns and trends in the Books Dataset.
- To compare book prices, price categories, and ratings.

- To explore the relationship between book prices and ratings.
- To create a dashboard that presents multiple visualizations together.
- To communicate findings through a clear and understandable data story.

3. Tools and Technologies

- **Python:** Programming language used to process and visualize data.
- **Pandas:** Used to load and organize the dataset.
- **Matplotlib:** Used to create charts, graphs, and the dashboard.
- **Visual Studio Code:** Used to write and execute the Python program.
- **CSV:** Used as the input data format.

4. Dataset Description

The dataset used in this task was books_dataset_5_pages.csv, created during Task 1. It contains book information collected from the first five pages of Books to Scrape. The dataset includes the title, price, availability, rating, and price category of each book.

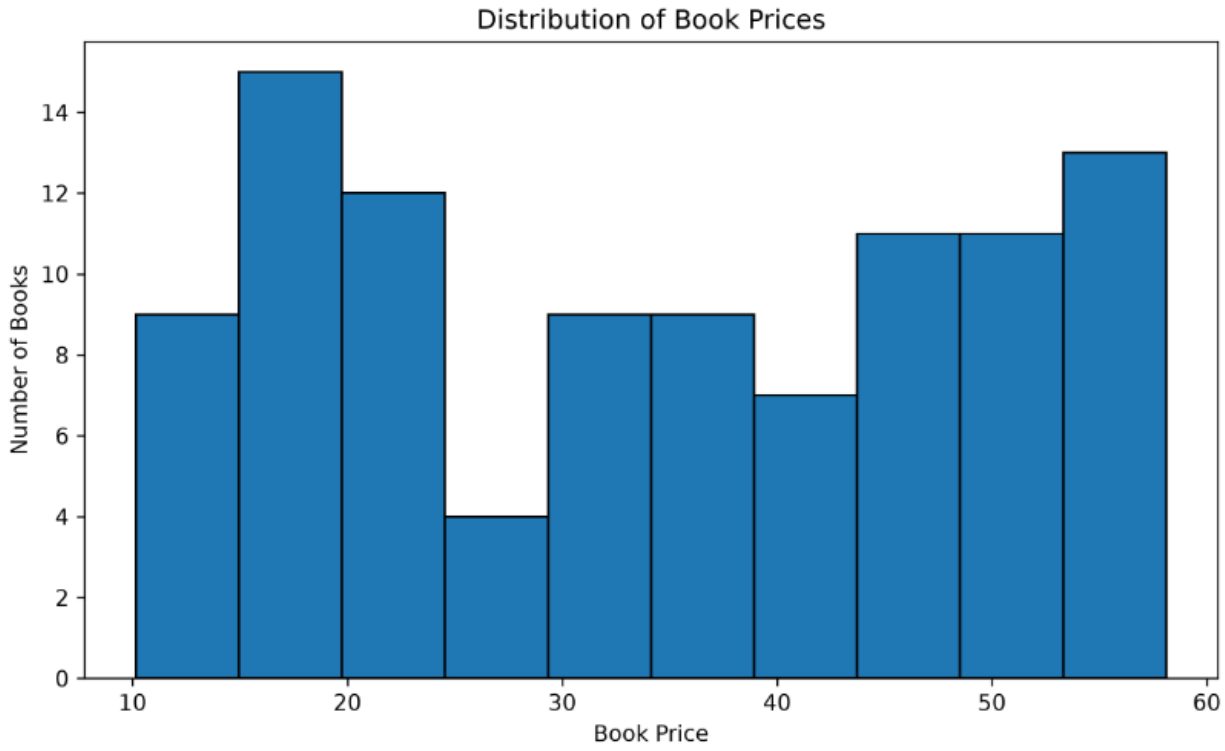
The dataset was reused to maintain consistency across the web scraping, exploratory data analysis, and data visualization tasks.

5. Visualization Methodology

The dataset was loaded into a Pandas DataFrame. The price column was converted into numeric values, and book ratings were mapped to numerical values from 1 to 5 for comparison. Four individual visualizations were created, followed by a dashboard combining all four charts.

6. Price Distribution Histogram

A histogram was created to show the distribution of book prices. It groups prices into intervals and displays the number of books within each interval. This visualization helps identify common price ranges and understand how prices are distributed.



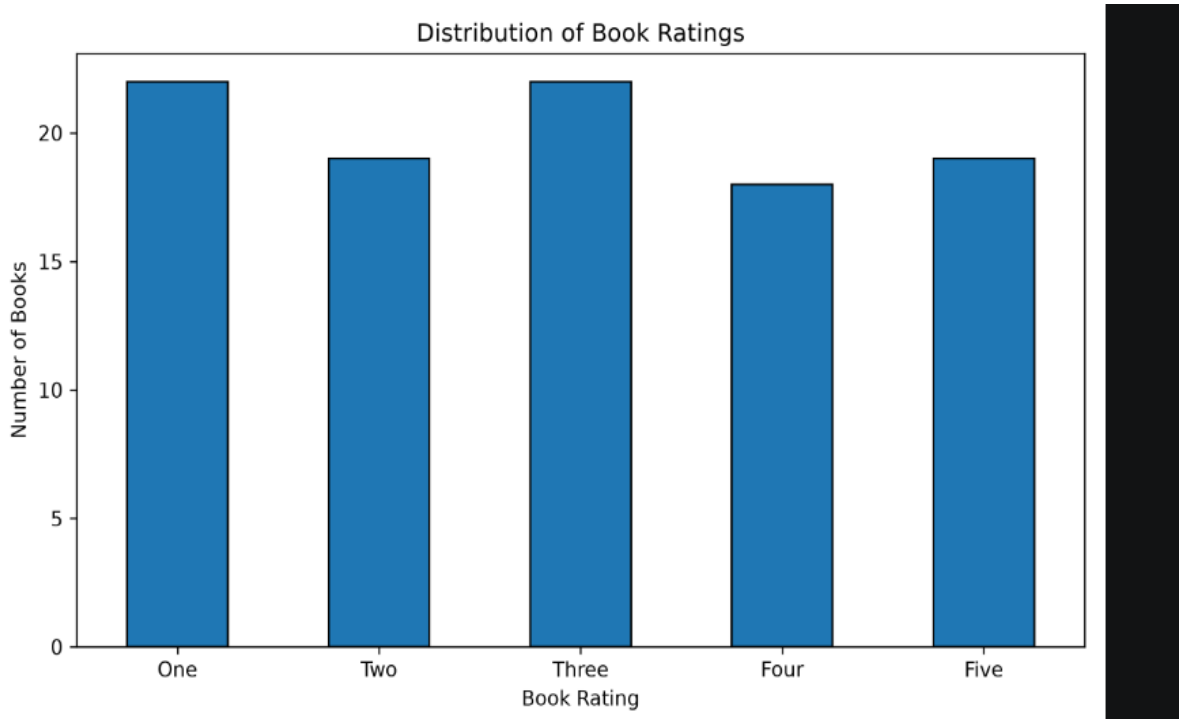
7. Price Category Comparison

A bar chart was created to compare the number of books in the Low, Medium, and High price categories. This visualization makes it easier to compare the size of each category and identify which category contains the most books.



8. Book Rating Distribution

A bar chart was developed to show the number of books receiving One, Two, Three, Four, and Five-star ratings. It provides a clear comparison of rating frequencies within the dataset.



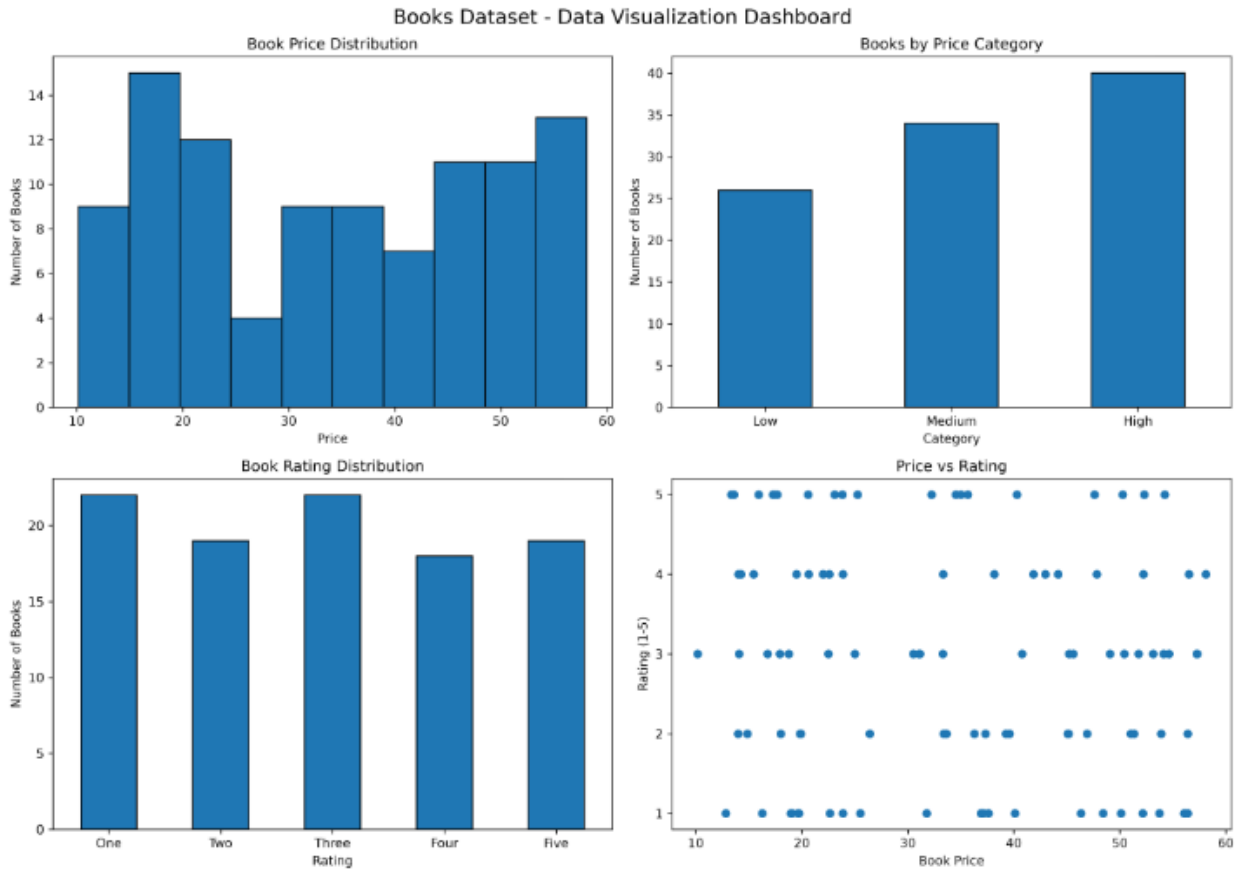
9. Relationship Between Price and Rating

A scatter plot was used to explore the relationship between book prices and ratings. Each point represents a book. The visualization helps identify possible patterns, although it does not by itself establish causation or statistical significance.



10. Data Visualization Dashboard

A dashboard was created by combining the price histogram, price category chart, rating distribution chart, and price-rating scatter plot into one figure. This provides an overview of several aspects of the dataset in a single view and makes comparisons easier.



11. Data Story and Key Findings

The visualizations were used to explore price distribution, category frequencies, rating patterns, and the relationship between price and rating. The charts help summarize the dataset and make comparisons easier than examining individual rows.

The final findings should be based on the actual charts and printed numerical results. The scatter plot should be interpreted cautiously because a visible pattern does not necessarily indicate a causal relationship.

12. Results

The task successfully produced four individual charts and a combined visualization dashboard using Python and Matplotlib. The generated images were saved as PNG files for inclusion in the report and future portfolio use.

13. Conclusion

This task demonstrated how raw data can be transformed into meaningful visual representations. Histograms, bar charts, scatter plots, and a combined dashboard were used to communicate information about book prices and ratings. The resulting visualizations support clearer data

interpretation and demonstrate the practical use of Python libraries for data analysis and presentation.